De La Salle University

SEMPHY01 — Semiconductor Physics

Long Assessment 1

From Bloch Bands to Equilibrium Semiconductor Carrier Statistics

Student:	TODO: Student Name
Student ID:	TODO: Student ID
Selected material:	TODO: Material and phase/polytype
Submission date:	TODO: Submission date
Computational record:	https://example.invalid/replace-me
PhysKit version:	TODO: PhysKit version/commit

This report concerns equilibrium carrier statistics only. Transport, mobility, conductivity, device behavior, and spatial band bending are outside the model unless explicitly identified as excluded context.

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1 Project declaration and scope

Table 1: Declared scope of the material model.

Item	Declaration
Material identity	**TODO:** Chemical formula and exact phase/polytype.
Sample model	**TODO:** Ideal bulk model and temperature range.
Gap model	**TODO:** Direct/indirect character and $E_g(T)$ representation.
Conduction edge	**TODO:** Location, equivalent valleys, and mass model.
Valence edge	**TODO:** Relevant bands, degeneracies, and mass model.
Statistics	**TODO:** Full Fermi–Dirac, Boltzmann comparison, or both.
n-type model	**TODO:** Donor-like species/defect and charge states.
p-type model	**TODO:** Acceptor-like species/defect and charge states.
Exclusions	**TODO:** Omitted physics and why it is outside scope.

1.1 Research question and modeling strategy

TODO: State the material-specific question investigated and summarize the path from the Bloch calculation to intrinsic and doped equilibrium statistics.

1.2 Use of PhysKit and student-authored work

The complete PhysKit library may be used [1]. Distinguish maintained package capabilities from your own configuration, analysis, extension, and interpretation.

Table 2: Computational capability and contribution map.

Capability	Source	Configuration, extension, or validation performed
Bloch Hamiltonian	PhysKit	**TODO:** Version, model, parameters, and checks.
DOS/state counting	**TODO:** Source	**TODO:** Normalization and validation.
Carrier statistics	**TODO:** Source	**TODO:** Integration/root-solving choices.
Dopant model	**TODO:** Source	**TODO:** Charge-state convention and extension.
Visualization	**TODO:** Source	**TODO:** How plots consume calculated results.

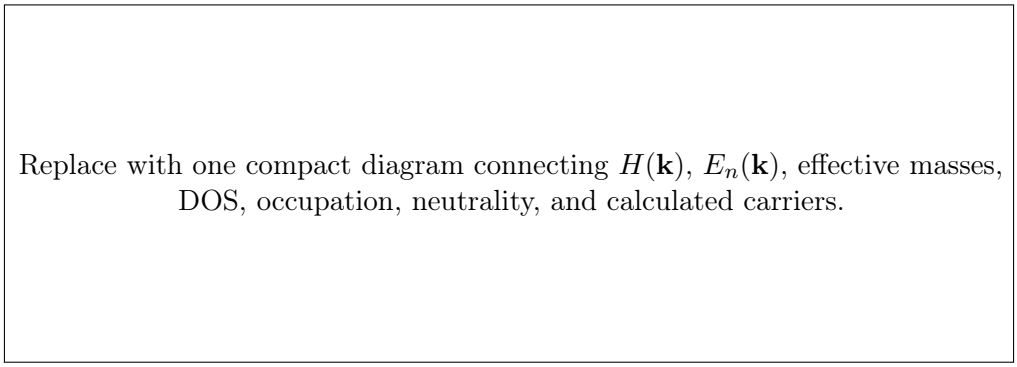


Figure 1: Replace with one compact diagram connecting $H(\mathbf{k})$, $E_n(\mathbf{k})$, effective masses, DOS, occupation, neutrality, and calculated carriers.

1.3 Model map

2 Part I: Bloch2D effective-mass reduction

2.1 Parent model and numerical representation

TODO: Describe the direct/reciprocal lattice, cell, boundary conditions, potential, representation, reciprocal cutoff or discretization, and units. Explain what PhysKit supplies and what you configured.

A useful operator statement is

$$H_{\mathbf{G}\mathbf{G}'}(\mathbf{k}) = \frac{\hbar^2|\mathbf{k}+\mathbf{G}|^2}{2m}\delta_{\mathbf{G}\mathbf{G}'} + V_{\mathbf{G}-\mathbf{G}'}. \tag{1}$$

TODO: Report a Hermiticity or equivalent operator-consistency check.

2.2 Band path and full-zone calculation

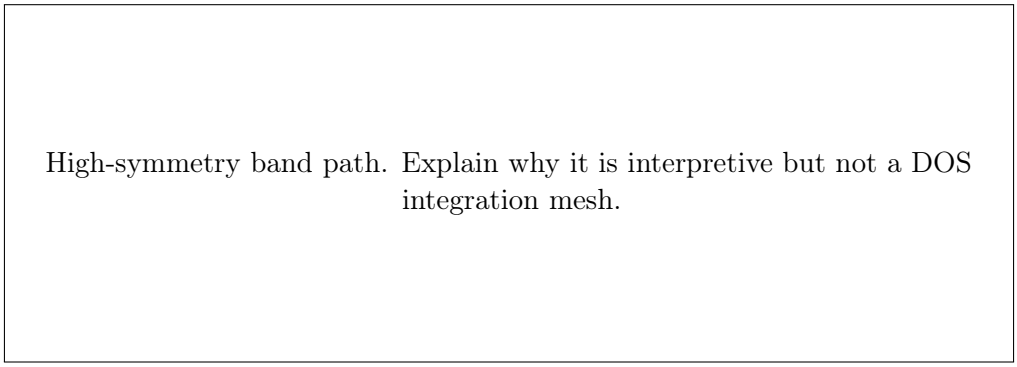


Figure 2: High-symmetry band path. Explain why it is interpretive but not a DOS integration mesh.

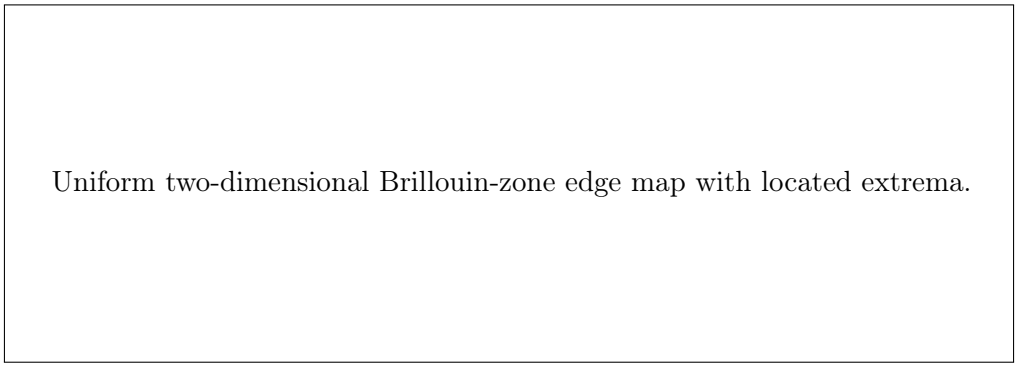


Figure 3: Uniform two-dimensional Brillouin-zone edge map with located extrema.

Table 3: Located band edges and fit domains.

Edge	Band/sheet	k_x	k_y	Energy	Classification
Valence maximum	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**
Conduction minimum	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**

2.3 Band edges and local quadratic model

For a conduction minimum,

$$E_c(\mathbf{k}_c + \mathbf{q}) \simeq E_{\mathrm{C}} + \frac{\hbar^2}{2}\mathbf{q}^{\mathsf{T}}\mathbf{M}_c^{-1}\mathbf{q}, \tag{2}$$

with the sign reversed before defining a positive hole-mass tensor at a valence maximum.

TODO: Describe the multidimensional fit, energy/radius window, tensor components, principal masses, principal directions, and any degeneracy or band folding that complicates the fit.

Table 4: Effective-mass tensor results.

Edge	$(M^{-1})_{xx}$	$(M^{-1})_{xy}$	$(M^{-1})_{yy}$	m_1/m_0	m_2/m_0	Fit window
Valence	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**
Conduction	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**

2.4 Convergence, residuals, and range of validity

TODO: Separate representation error, **k**-mesh error, fit-window error, and physical nonparabolicity. State a bounded energy range over which the parabolic approximation is adequate.

2.5 Two-dimensional DOS and state counting

For one anisotropic parabolic valley,

$$m_{d,2D} = \sqrt{m_1 m_2}, \qquad g_{2D}(E) = \frac{g_s g_v m_{d,2D}}{2\pi\hbar^2}\Theta(E - E_{\mathrm{C}}). \tag{3}$$

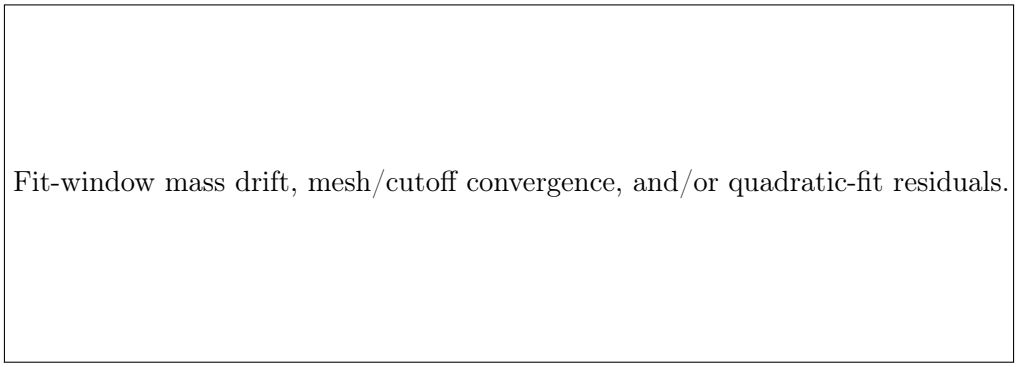


Figure 4: Fit-window mass drift, mesh/cutoff convergence, and/or quadratic-fit residuals.

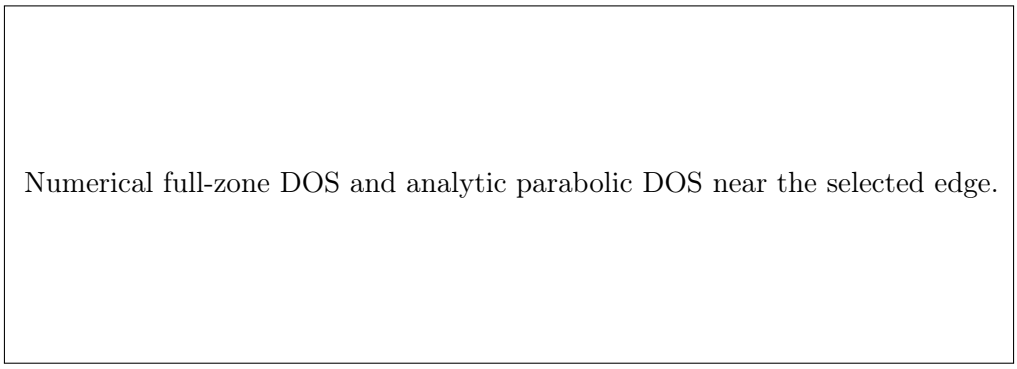


Figure 5: Numerical full-zone DOS and analytic parabolic DOS near the selected edge.

TODO: Verify integrated state counts and discuss mesh spacing, broadening, finite bandwidth, additional valleys/sheets, and nonparabolicity. Emphasize that these are two-dimensional sheet quantities.

3 Part II: Intrinsic three-dimensional model

3.1 Band-edge state model and DOS components

TODO: Define the energy reference, gap, all conduction valleys and valence bands, principal masses, and degeneracies. Explain whether quoted DOS masses already include valley multiplicity.

For each three-dimensional parabolic component,

$$g_c(E) = \frac{g_s g_v}{4\pi^2}\left(\frac{2}{\hbar^2}\right)^{3/2}\sqrt{\det \mathbf{M}_c}\sqrt{E - E_{\mathrm{C}}}\,\Theta(E - E_{\mathrm{C}}), \tag{4}$$

with an analogous expression below E_{V} for holes.

Table 5: Components of the effective densities of states at a representative temperature.

Component	g_s	g_v	m_1/m_0	m_2/m_0	m_3/m_0	Contribution
TODO: Conduction valley	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**
TODO: Valence band	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**

3.2 Fermi–Dirac occupation and carrier integrals

$$f(E,\mu,T) = \frac{1}{1+\exp[(E-\mu)/(k_{\mathrm{B}}T)]}, \tag{5}$$

$$n(\mu,T) = \int_{E_{\mathrm{C}}}^{\infty} g_c(E) f(E,\mu,T)\, dE, \tag{6}$$

$$p(\mu,T) = \int_{-\infty}^{E_{\mathrm{V}}} g_v(E)[1-f(E,\mu,T)]\, dE. \tag{7}$$

TODO: Explain stable evaluation, units, integration method, cutoffs, tolerances, and convergence evidence.

3.3 Intrinsic neutrality and nondegenerate comparison

$$n(\mu_{\mathrm{i}},T) - p(\mu_{\mathrm{i}},T) = 0. \tag{8}$$

TODO: Report the root bracket, tolerance, residual, and temperature sweep.

$$n_{\mathrm{i}} = \sqrt{N_{\mathrm{C}}N_{\mathrm{V}}}\exp\left(-\frac{E_{\mathrm{g}}}{2k_{\mathrm{B}}T}\right), \tag{9}$$

$$\mu_{\mathrm{i}} = \frac{E_{\mathrm{C}}+E_{\mathrm{V}}}{2} + \frac{k_{\mathrm{B}}T}{2}\ln\left(\frac{N_{\mathrm{V}}}{N_{\mathrm{C}}}\right). \tag{10}$$

Three-dimensional DOS and occupation at one temperature.

Figure 6: Three-dimensional DOS and occupation at one temperature.

TODO: Compare full Fermi–Dirac integration with the nondegenerate limit and identify when they agree.

3.4 Silicon control and validation evidence

TODO: Summarize automated or reproducible checks for monotonicity, energy-shift invariance, isotropic/anisotropic limits, $T^{3/2}$ scaling, neutrality, and integration convergence.

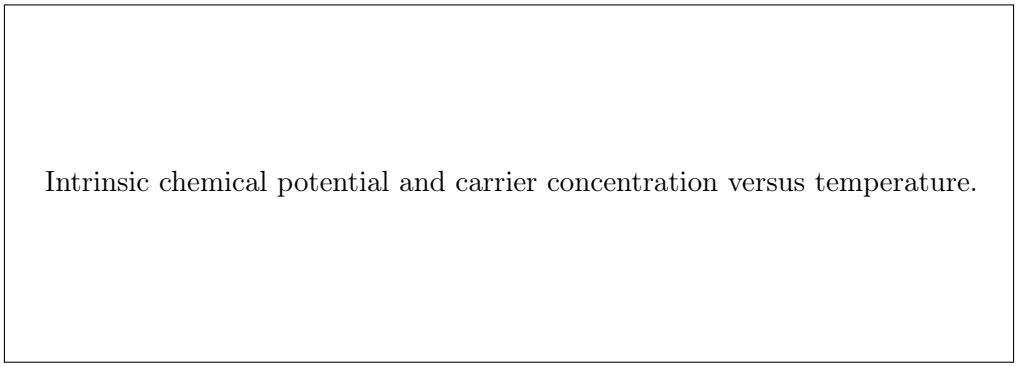


Figure 7: Intrinsic chemical potential and carrier concentration versus temperature.

Table 6: Silicon control at $3.00 \times 10^2\,\mathrm{K}$.

Quantity	Input/result	Reference value	Difference	Interpretation
E_g	**TODO:**	approximately 1.12 eV	**TODO:**	**TODO:**
n_i	**TODO:**	approximately $1.5 \times 10^{10}\,1/\mathrm{cm}^3$	**TODO:**	**TODO:**
μ_i	**TODO:**	**TODO:** source-dependent	**TODO:**	**TODO:**

4 Part III: Literature-based material reconstruction

4.1 Source strategy and parameter provenance

TODO: Explain how curated sources, primary literature, reviews, and published band information were reconciled. Do not mix phases or definitions silently.

Table 7: Material-parameter provenance.

Quantity	Symbol	Value/model	Unit	Temperature	Definition	Source
Band gap	E_g	**TODO:**	eV	**TODO:**	direct/indirect	**TODO:** cite
Electron mass	$\mathbf{M}_c$	**TODO:**	m_0	**TODO:**	curvature/DOS	**TODO:** cite
Hole mass	$\mathbf{M}_v$	**TODO:**	m_0	**TODO:**	HH/LH/DOS	**TODO:** cite
Valley count	g_v	**TODO:**	1	**TODO:**	included in mass?	**TODO:** cite
Published N_C	N_C	**TODO:**	cm^{-3}	**TODO:**	measured/derived	**TODO:** cite
Published N_V	N_V	**TODO:**	cm^{-3}	**TODO:**	measured/derived	**TODO:** cite
Published n_i	n_i	**TODO:**	cm^{-3}	**TODO:**	method	**TODO:** cite

4.2 Band-diagram interpretation

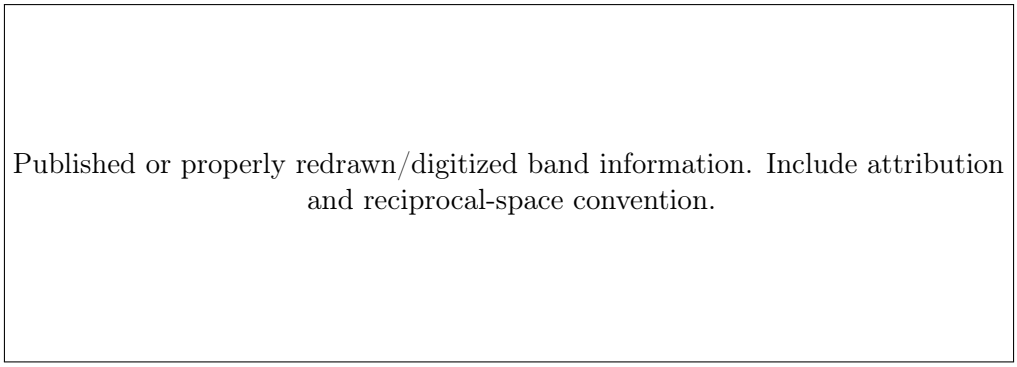


Figure 8: Published or properly redrawn/digitized band information. Include attribution and reciprocal-space convention.

TODO: Identify edge locations, direct/indirect character, multiplicity, nearby bands, anisotropy, nonparabolicity, and limitations of the plotted path.

4.3 Material model and intrinsic results

TODO: Define $E_g(T)$, modeled components, temperature range, and any material-specific use or extension of PhysKit.

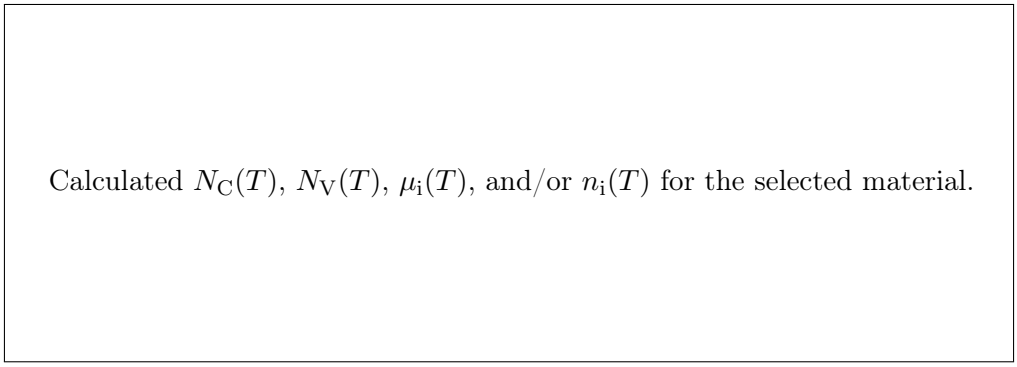


Figure 9: Calculated $N_C(T)$, $N_V(T)$, $\mu_i(T)$, and/or $n_i(T)$ for the selected material.

4.4 Published comparison and sensitivity

Table 8: Comparison with independent published data.

Quantity	T	Calculated	Published	Unit	Difference	Source
TODO:	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:** cite

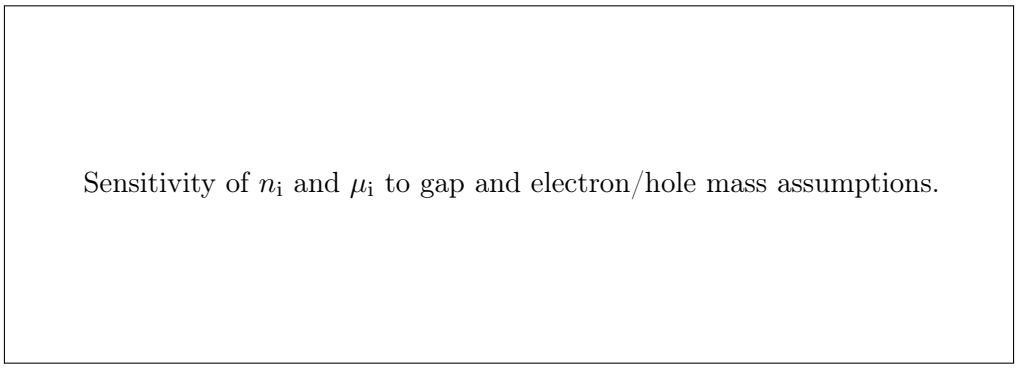


Figure 10: Sensitivity of n_i and μ_i to gap and electron/hole mass assumptions.

TODO: Separate numerical error, parameter uncertainty, and physical model limitations when interpreting disagreement.

Table 9: Dopant or defect-level provenance.

Kind	Species/site	Level	Charge states	Degeneracy convention	Range	Source/limitations
Donor	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:** cite
Acceptor	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:** cite

5 Part IV: Equilibrium doping

5.1 Donor and acceptor models

With the explicitly declared degeneracy convention, one possible model is

$$N_D^+ = \frac{N_D}{1 + g_D \exp[(\mu - E_D)/(k_\mathrm{B}T)]}, \tag{11}$$

$$N_A^- = \frac{N_A}{1 + g_A \exp[(E_A - \mu)/(k_\mathrm{B}T)]}, \tag{12}$$

$$0 = p + N_D^+ - n - N_A^-. \tag{13}$$

TODO: Explain the convention, low/high-temperature behavior, solver bracket, and neutrality residual.

5.2 Representative intrinsic, n-type, p-type, and compensated cases

Table 10: Modeled equilibrium cases.

Case	N_D	N_A	Temperature range	Physical purpose
Intrinsic	0	0	**TODO:**	baseline
n-type	**TODO:**	**TODO:**	**TODO:**	**TODO:**
compensated n-type	**TODO:**	**TODO:**	**TODO:**	**TODO:**
p-type	**TODO:**	**TODO:**	**TODO:**	**TODO:**
compensated p-type	**TODO:**	**TODO:**	**TODO:**	**TODO:**

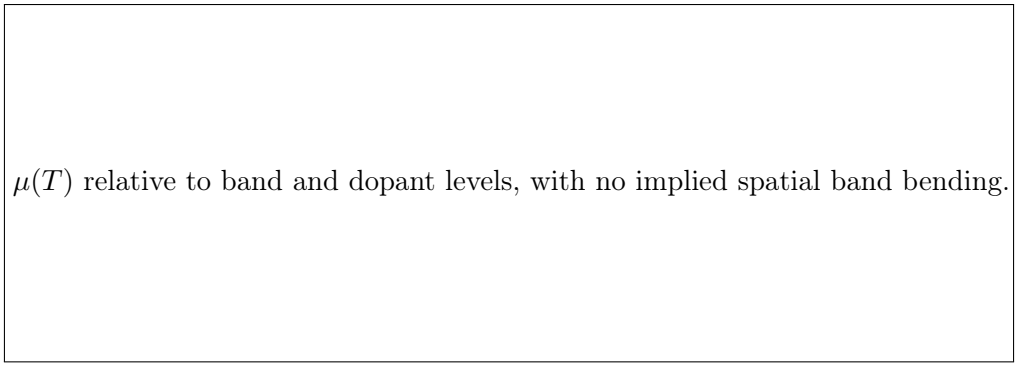


Figure 11: $\mu(T)$ relative to band and dopant levels, with no implied spatial band bending.

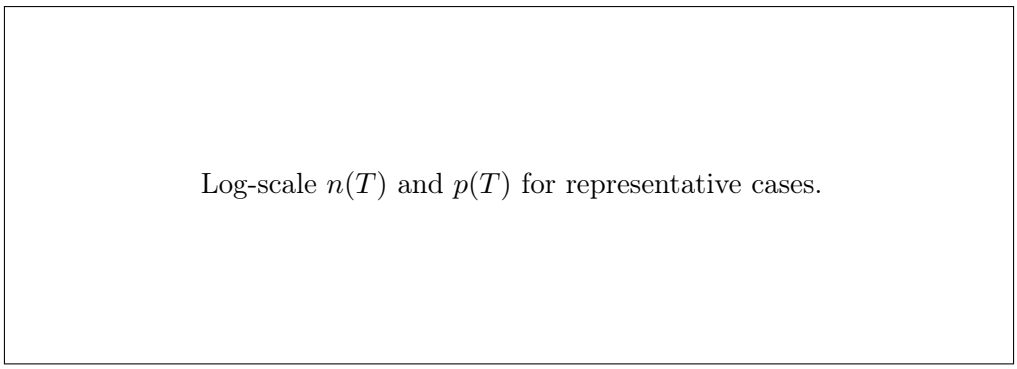


Figure 12: Log-scale $n(T)$ and $p(T)$ for representative cases.

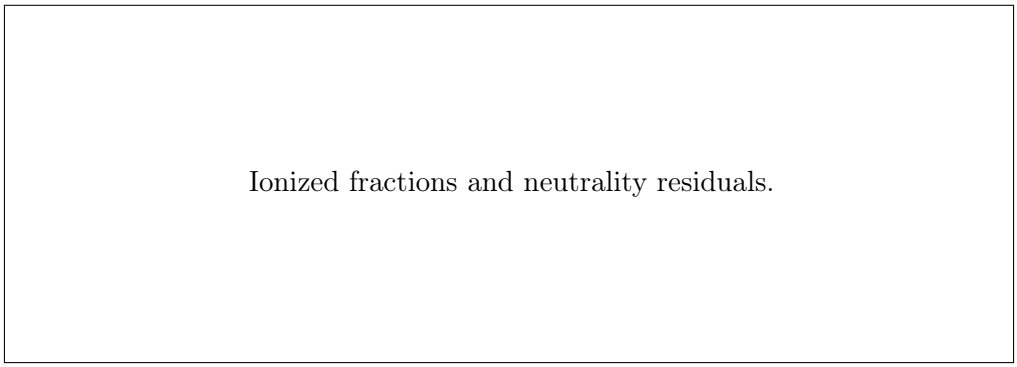


Figure 13: Ionized fractions and neutrality residuals.

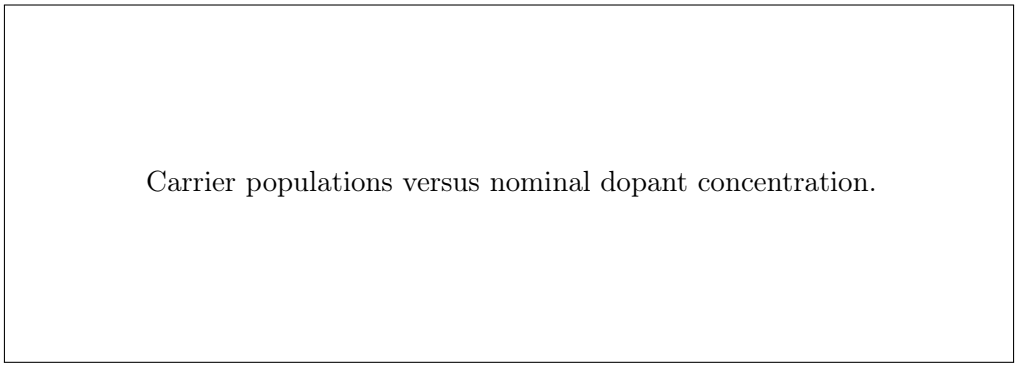


Figure 14: Carrier populations versus nominal dopant concentration.

5.3 Approximation comparison and regimes

TODO: Identify freeze-out, extrinsic, intrinsic, and degenerate/non-Boltzmann regimes where physically accessible. Compare incomplete and full ionization, and state where the simpler result differs by less than five percent.

6 Synthesis and bounded validity

6.1 What transfers from Bloch2D to the real material?

TODO: Separate transferable numerical ideas—curvature fitting, state counting, occupation, convergence—from dimensionality and material-specific physics that do not transfer directly.

6.2 Adequacy of the final model

TODO: Answer quantitatively: over what temperature and concentration range is the effective-mass plus equilibrium dopant model adequate, and which omitted physical effect is most likely to invalidate it first?

6.3 Main conclusion

TODO: Give a concise answer supported by your most decisive table and figure.

7 Reproducibility and validation record

Table 11: Computational environment and execution record.

Item	Value
Repository/archive	`https://example.invalid/replace-me`
Python version	**TODO:**
PhysKit version/commit	TODO: PhysKit version/commit
Main dependencies	**TODO:**
Run order	**TODO:**
Random seeds, if any	**TODO:**
Clean-run result	**TODO:**
Test/validation summary	**TODO:**

TODO: State how the submitted figures and tables can be regenerated. Disclose materially reused code, data, and computational assistance according to course policy.

References

[1] Eugene Joseph M. Ragasa. Project koios physkit, 2026. Use the exact version or commit used in the submitted calculation.

A Supplementary convergence and validation evidence

TODO: Place useful evidence here when it would interrupt the main argument. Do not use the appendix to hide checks needed to support a main conclusion.

B Additional parameter and provenance records

TODO: Include extended tables, digitization details, or source-definition comparisons if needed.